MODULE 3 · MACHINE LEARNING · DAY 33

Drawing the Best Line

Your first machine learning model — simple linear regression.

5	17	2
sections	worked pictures	numbers to find



Boston University

Prepared & presented by **Srinivasa Sai Chava**

WHERE WE ARE

Today you build a model, not just talk about one

1 **Day 31**

What a model IS. Features, labels, and why we hide rows.

2 **Day 32**

Getting the columns into a shape the model can read.

3 **TODAY**

Build the simplest model that exists, from scratch, by hand.

Everything you need, you already have

No new maths. If you can read the equation of a straight line from school, you can follow every slide today. We will do the whole thing on FIVE students first, with numbers small enough to check in your head — and only then switch on the computer.

A straight line is a toy model. But every idea in it — error, best fit, R^2, overfitting — comes back unchanged in every model for the rest of the course.

SECTION 1 OF 5

What a line is for

Five students, one question, and the rule that answers it.

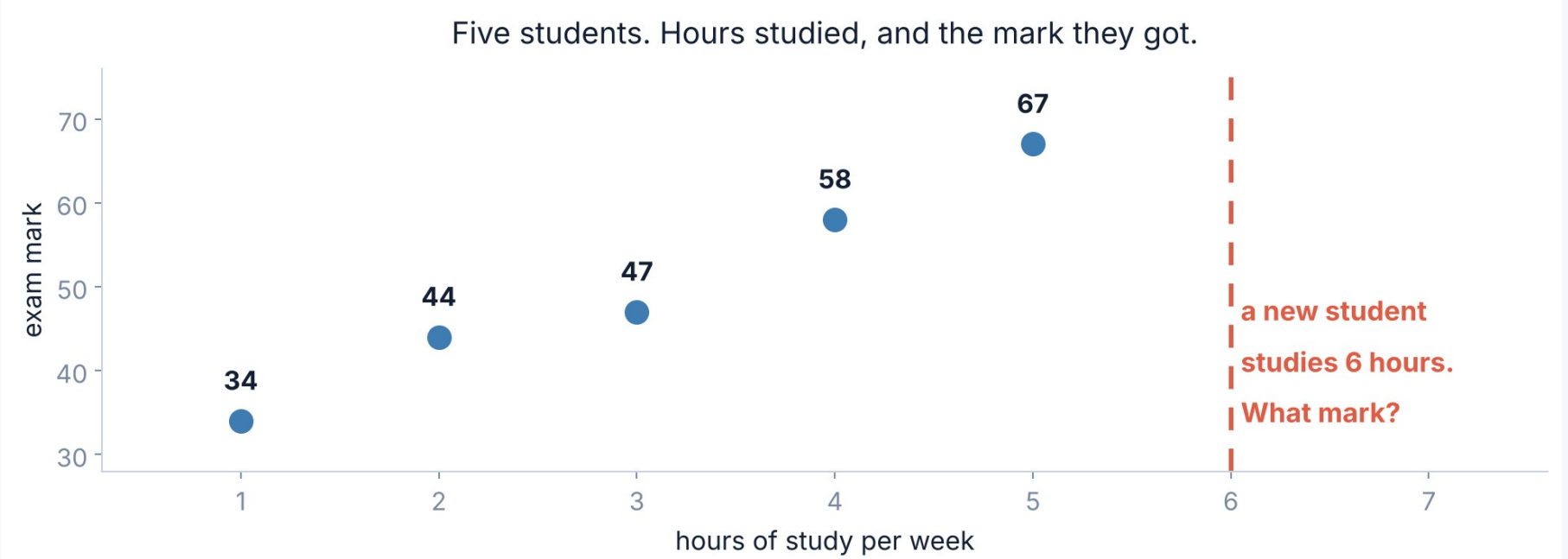
- A new student, no data
- What a line really is
- Slope and intercept
- Which line is best?

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THE PROBLEM

Five students. One question.

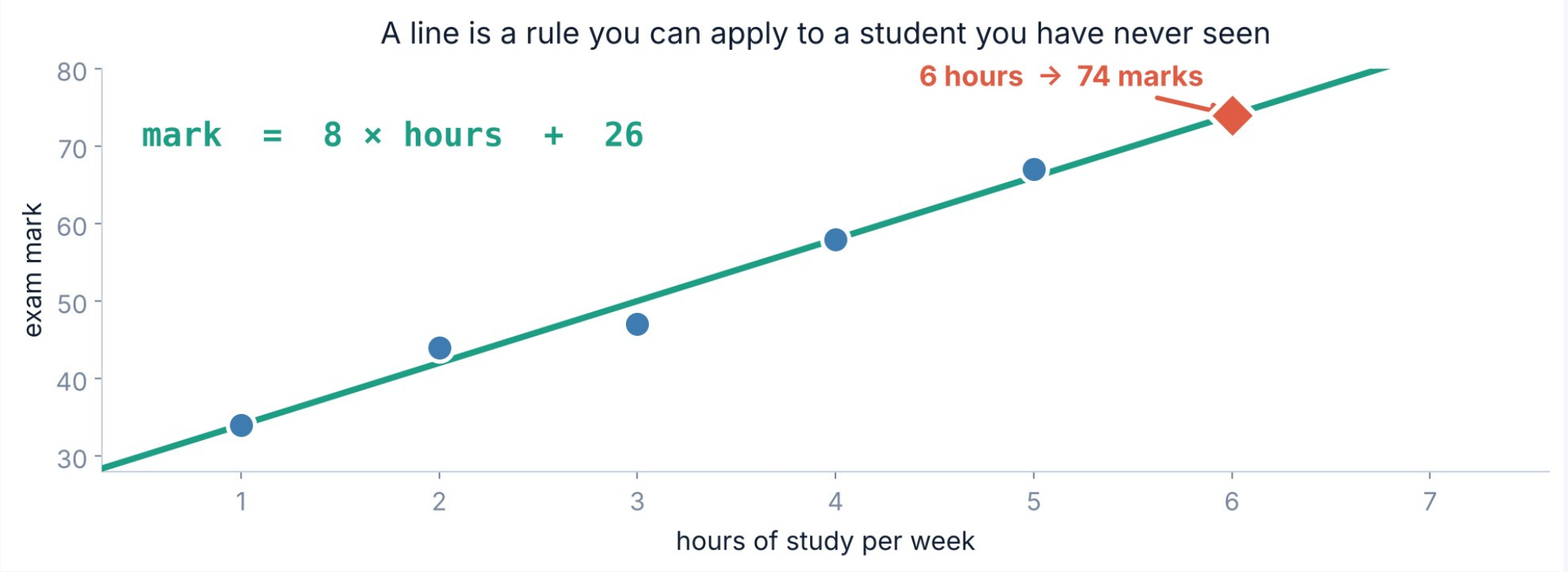
EXAMPLE **You tracked five classmates: how many hours a week they studied, and what they scored.**



You cannot look this student up in the table — she is not in it. You need a RULE that turns any number of hours into a mark.

THE IDEA

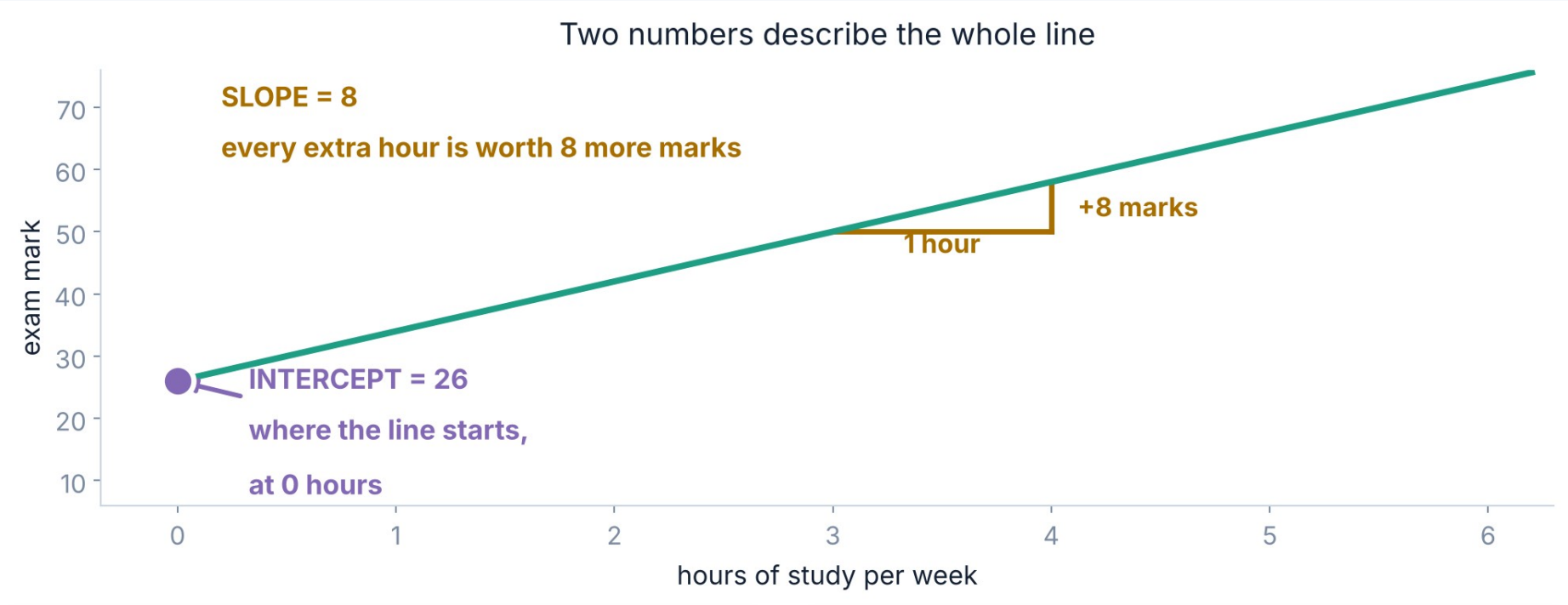
A line is a rule you can carry around



Give the rule any number of hours and it gives you back a mark. It works for students who were never in your table. That is the whole point of a model.

SLOPE AND INTERCEPT

The whole model is two numbers



SLOPE = 8

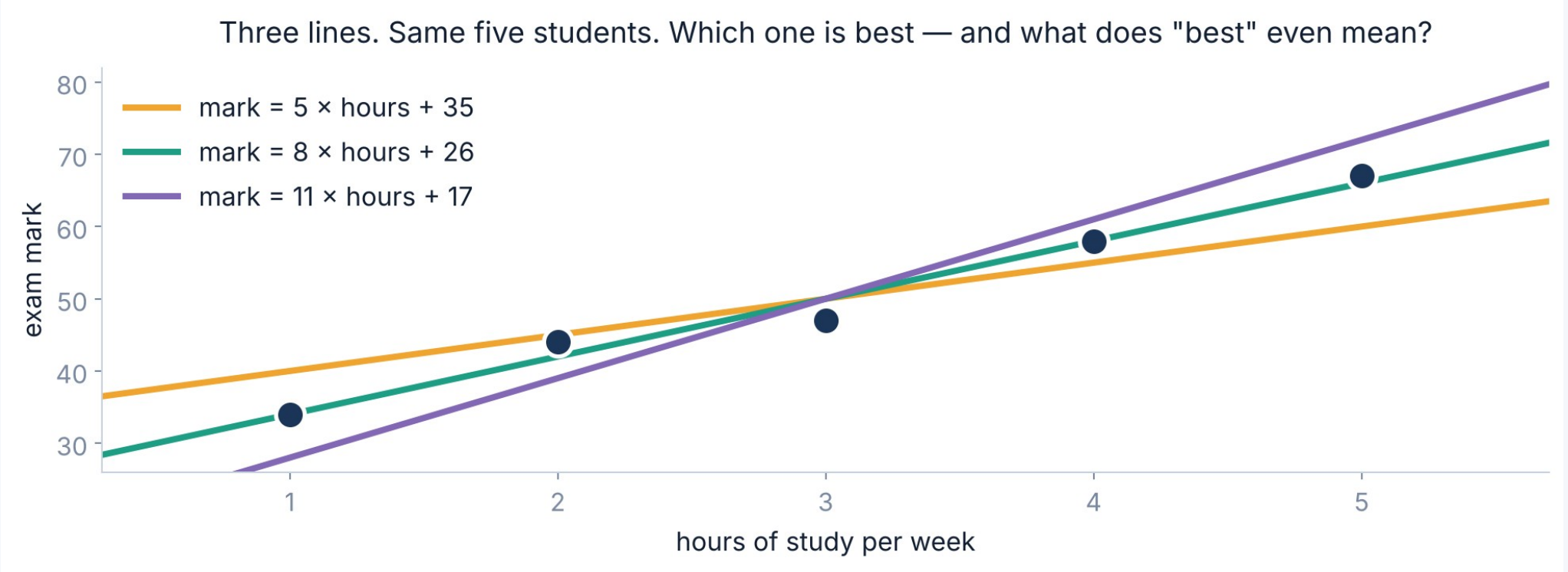
how much the mark goes up for one extra hour

INTERCEPT = 26

the mark the line predicts at zero hours

THE REAL QUESTION

But I could draw a hundred lines. Which one?



All three go through the middle of the dots. Before we can pick one, we need to agree on what "best" MEANS — as a number we can compute.

SECTION 2 OF 5

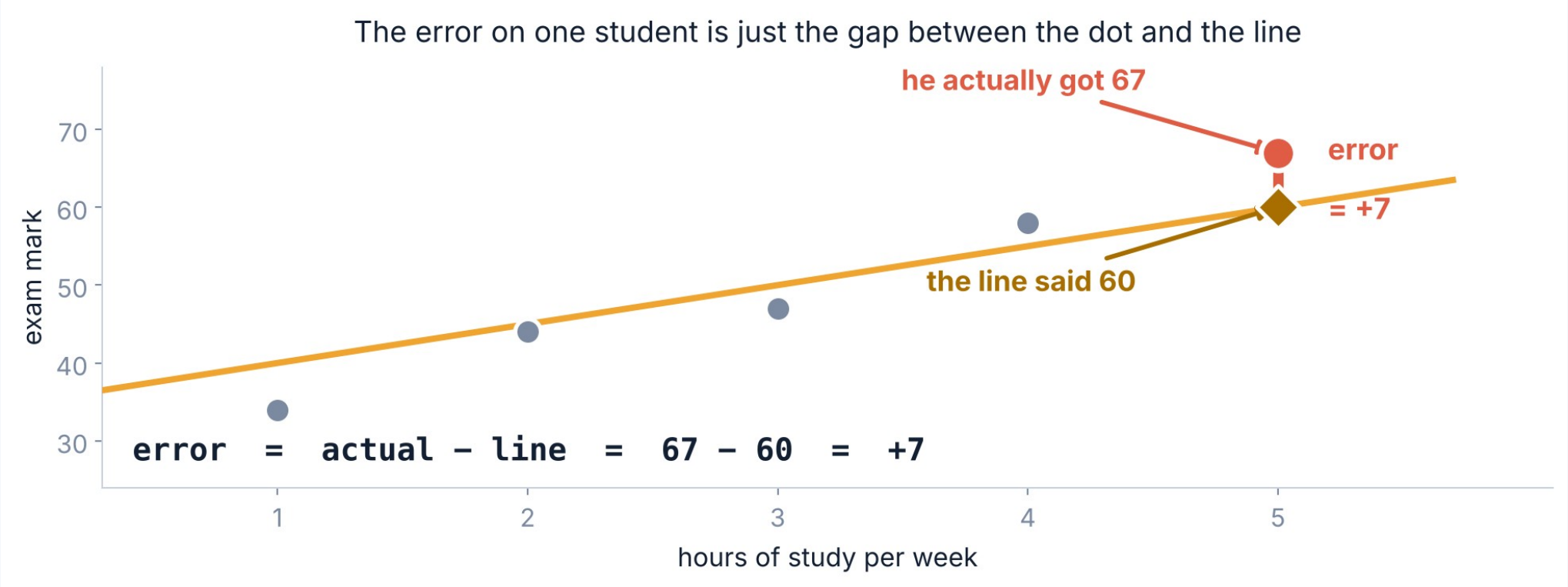
How wrong is a line?

Turn "best" from an opinion into a number.

- The error on one student
- Why adding fails
- Squaring
- Least squares

MEASURING THE DAMAGE

Start with one student



One number per student: how far the dot sits from the line. Positive means the line guessed too low. That is all an error is.

THE FIRST IDEA, AND WHY IT FAILS

Adding the errors up does not work



The best line and the exactly-upside-down line BOTH score zero. The pluses and the minuses cancel, and the score tells you nothing at all.

THE FIX

So square every error before adding

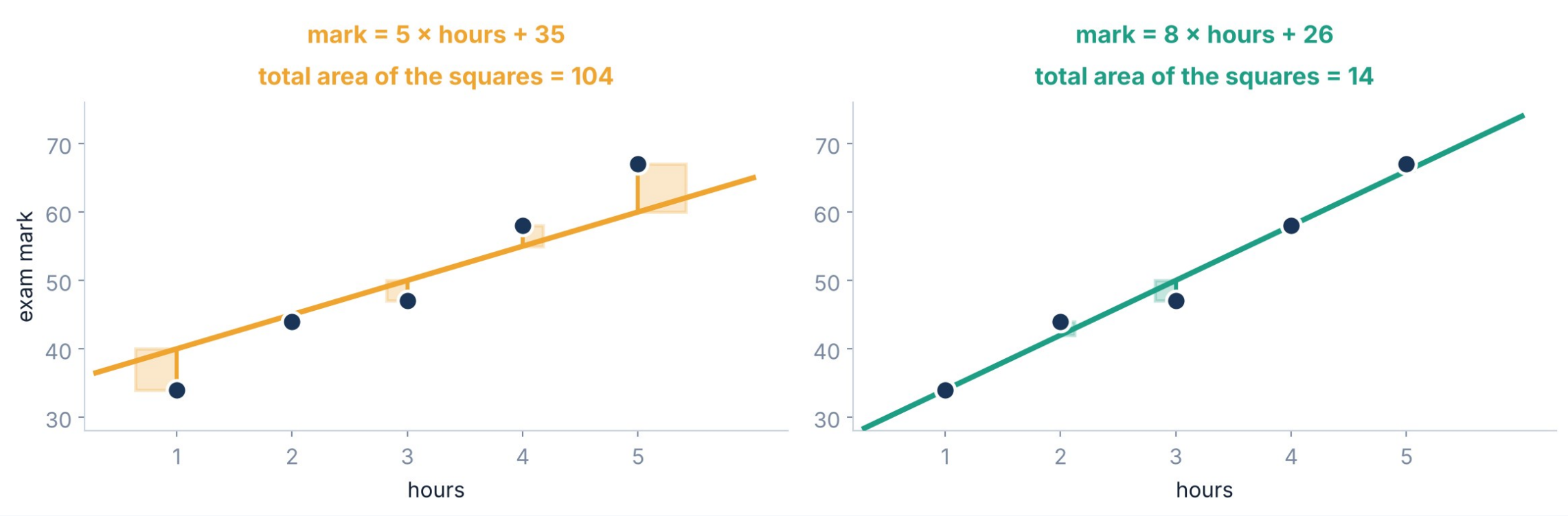
hours	actual mark	the line says	error	error squared
1	34	34	0	0
2	44	42	+2	4
3	47	50	−3	9
4	58	58	0	0
5	67	66	+1	1
		total	**0**	**14**

Squaring throws away the sign. A miss of −3 and a miss of +3 both cost 9.

Total squared error = 14. Now "best" has a definition: the best line is the one with the SMALLEST total squared error. That name is literally "least squares".

THE PICTURE BEHIND THE NAME

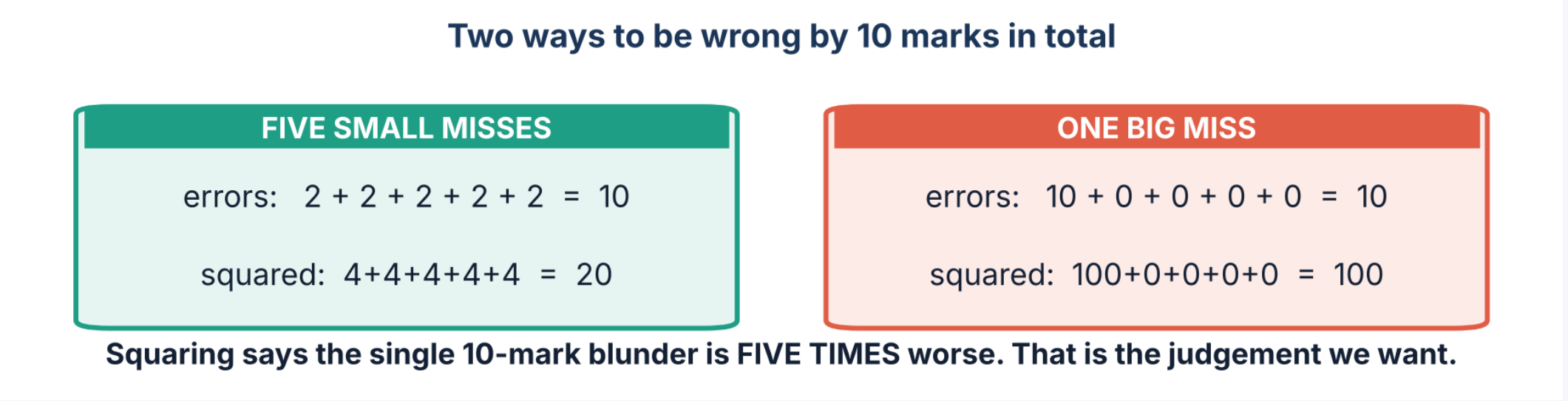
Why it is called least squares



Each error becomes an actual square on the chart. Fitting a line means shrinking the total shaded area — from 104 down to 14.

A FAIR QUESTION

Why squaring, and not just ignoring the minus sign?



Both options remove the sign. Squaring goes further: it says a few small misses are forgivable, but one enormous miss is not. For predicting a mark, a model that is usually within 2 and occasionally out by 30 is worse than one that is always out by 5 — and squaring is what encodes that preference.

SECTION 3 OF 5

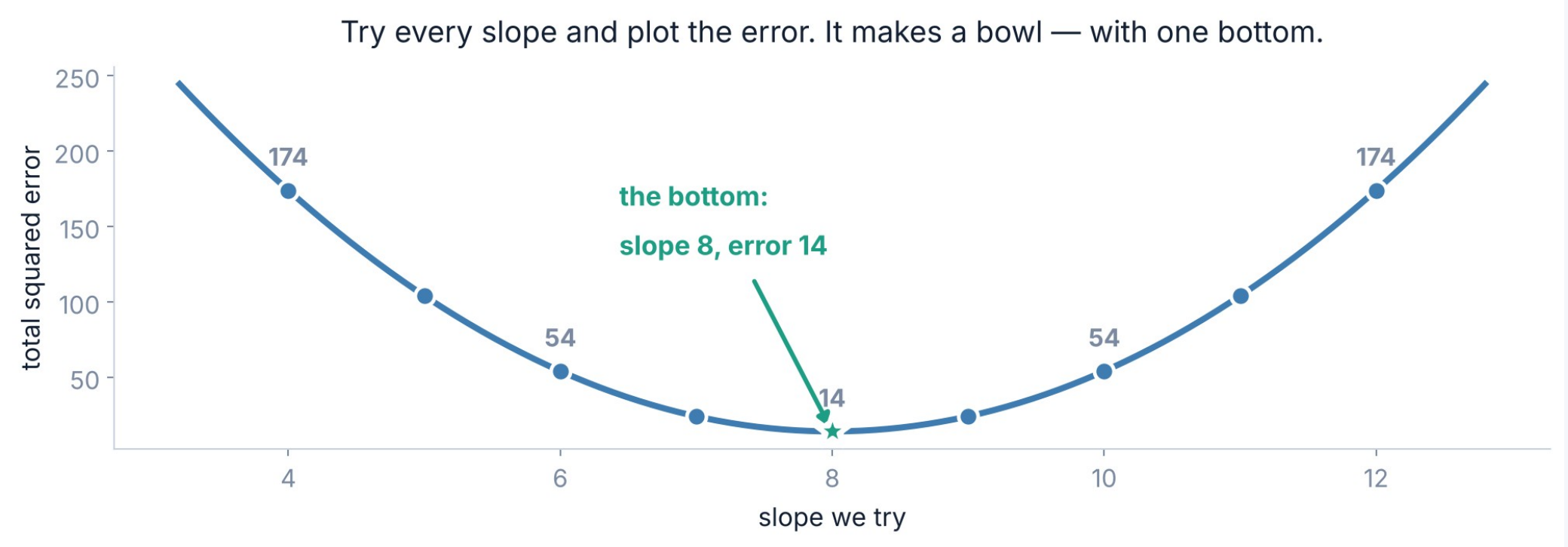
Finding the best one

Two different routes. Exactly the same answer.

- The bowl
- The formula, by hand
- Gradient descent
- Why both matter

WHAT WE ARE LOOKING FOR

Try every slope and the errors make a bowl



There is exactly one bottom, and no other dips to get stuck in. For a straight line this is always true — which is why the problem is solvable at all.

THE EXACT ANSWER

Route 1 · There is a formula

$$\text{slope} = \frac{\text{sum of (hours} - \text{average hours)} \times \text{(mark} - \text{average mark)}}{\text{sum of (hours} - \text{average hours)}^2}$$

In words:

- For every student, work out how far they sit from the average hours, and from the average mark.
- Multiply those two gaps together. A student above average in both pushes the slope UP.
- Add it all up, then divide by how spread out the hours were.

You will never type this formula in real life — one line of scikit-learn does it. Work through it once so that the answer is never magic.

BY HAND

The same formula, worked on our five students

hours	mark	hours − 3	mark − 50	multiply	(hours − 3)²
1	34	−2	−16	32	4
2	44	−1	−6	6	1
3	47	0	−3	0	0
4	58	+1	+8	8	1
5	67	+2	+17	34	4
			total	**80**	**10**

$$80 \div 10 = 8$$

the slope

→

$$50 - 8 \times 3 = 26$$

the intercept

mark = 8 × hours + 26. The average student does 3 hours and gets 50, so the line is forced through that point — the intercept just follows.

GRADIENT DESCENT

Route 2 · Guess, then walk downhill

THE IDEA, IN FOUR LINES

- Start with any slope and intercept at all. Zero and zero will do.
- Work out the total squared error.
- Nudge both numbers in whatever direction makes that error smaller.
- Repeat a few hundred times, and stop when it stops improving.

WHY BOTHER, IF A FORMULA EXISTS?

Because the formula only exists for simple cases like a straight line. A neural network has millions of numbers to choose and no formula at all that could ever give them to you.

Walking downhill still works there. It is how essentially every modern model is trained — including the ones behind the chatbots you use.

Route 1 is a shortcut that happens to exist for straight lines. Route 2 is the method that scales to everything else — which is why it is worth meeting on a problem small enough to check by hand.

NO CODE YET

One step of gradient descent, done by hand

Start from the worst possible guess: slope 0, intercept 0 — a flat line along the bottom.

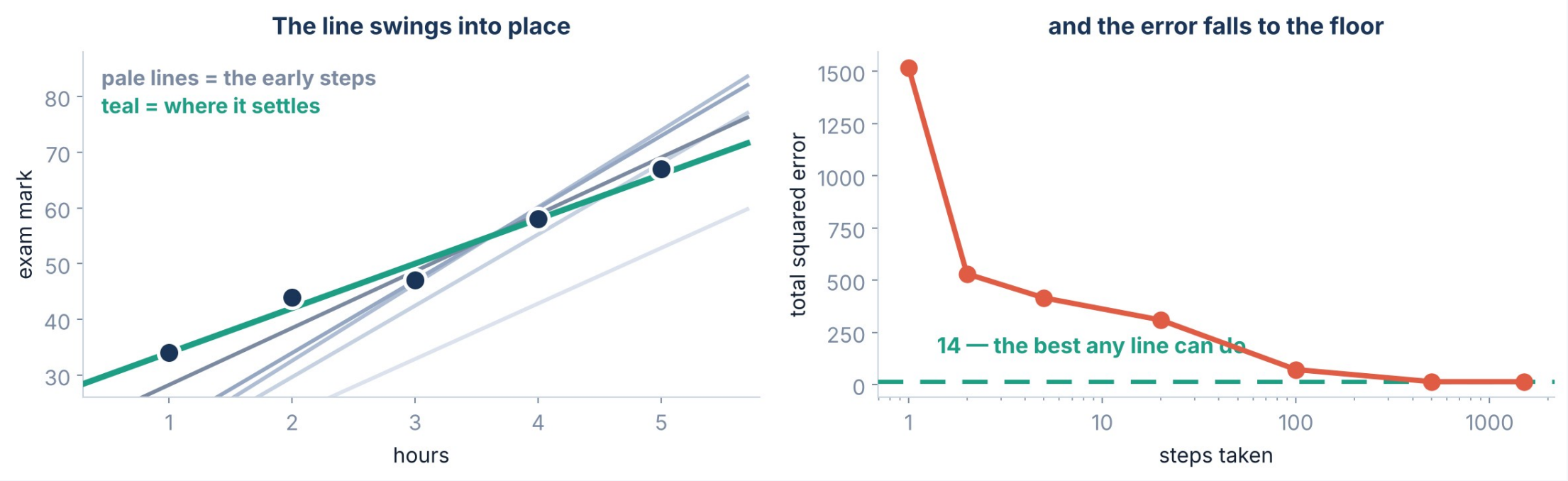
step	what you compute	on our numbers	result
1	every error (line − actual)	−34, −44, −47, −58, −67	all badly too low
2	slope nudge = ⅖ × Σ(error × hours)	⅖ × (−830)	−332
3	intercept nudge = ⅖ × Σ(error)	⅖ × (−250)	−100
4	step a little way against each nudge	0 − 0.03 × (−332)	slope → 9.96
5	and the same for the intercept	0 − 0.03 × (−100)	intercept → 3.00

That 0.03 is the LEARNING RATE — how big a step to take. Too small and it crawls; too big and it flies past the bottom.

One step took us from slope 0 to slope 9.96. The error fell from 13,154 to 1,518. Nothing clever happened — we just moved in the direction that helped.

GRADIENT DESCENT, RUNNING

Repeat it, and watch the line settle



after	1 step	2	5	20	100	500	1500
slope	9.96	12.81	13.79	12.99	10.21	8.04	8.00
intercept	3.00	4.03	5.00	7.98	18.03	25.86	26.00
total squared error	1518	530	415	309	72	14.0	14.0

Two routes.
The same two numbers.

The formula gets there in one calculation. Gradient descent takes about 800 steps. Both land on slope 8 and intercept 26 — because there is only one bottom to the bowl.

SECTION 4 OF 5

Reading the model

A line you cannot explain is not worth having.

- What the slope means
- The intercept trap
- R^2, honestly
- Looking at the errors

READING THE MODEL

Say the slope out loud, in English

$$\text{mark} = 8 \times \text{hours} + 26$$

THE SLOPE, 8

“One extra hour of study a week is worth about 8 more marks.”

This is the sentence you would actually say to a student. It is the whole reason anyone fits a line instead of a black box.

THE INTERCEPT, 26

“A student who studies zero hours would score 26.”

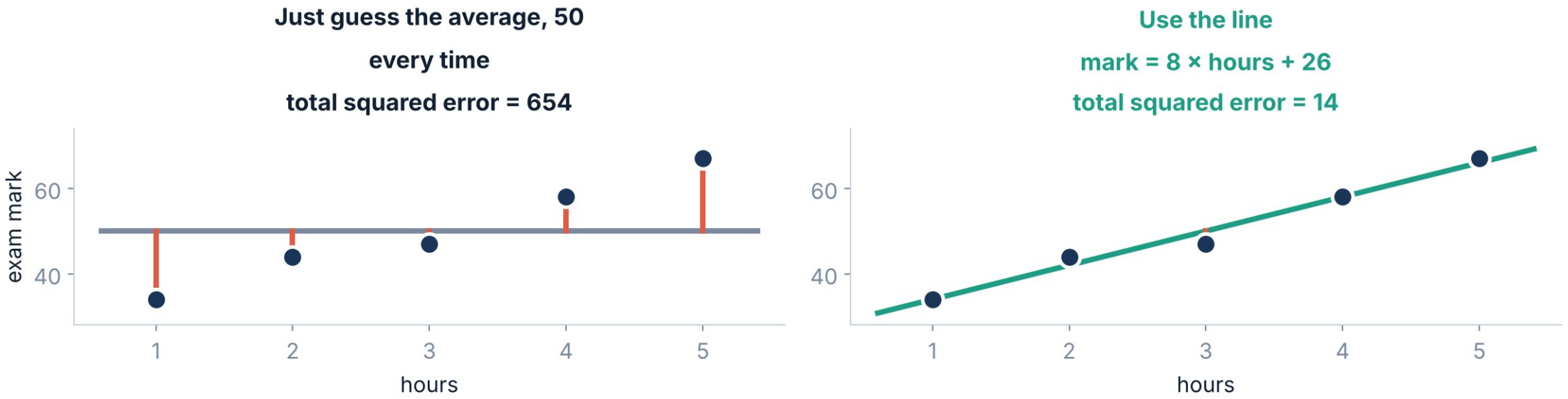
True of the LINE. But nobody in our five studied zero hours — the lowest was 1. So this number is an extrapolation, and you should not quote it.

If you cannot finish the sentence “one more unit of X is worth ___ of Y”, you have not understood your own model yet.

SCORING THE MODEL

R^2 · compared with just guessing the average

R^2 asks one question: how much better is the line than guessing the average?



$R^2 = 1 - (14 \div 654) = 0.979$ the line removed 97.9% of the error that guessing the average would have left

AND WHAT IT IS NOT

What R^2 is actually telling you

R^2	what it means	should you be happy?
1.00	every point sits exactly on the line	**No — check for a mistake**
0.98	our five students	Yes, but five points is not evidence
0.60	a real, noisy relationship	Often very useful
0.00	no better than always guessing the average	The line found nothing
below 0	worse than guessing the average	**Something is wrong**

A high R^2 does NOT mean the line is the right shape, that hours CAUSE marks, or that the model will work on new students. It only compares the line with the flat average.

A good R^2 is a reason to look closer, never a reason to stop looking. The next slide shows an R^2 of 0.75 hiding a line that is simply the wrong shape.

THE RESIDUAL PLOT

Always look at the errors, not just the score



Plot the errors against the input. If they scatter randomly around zero, the line fits. If they make a PATTERN, the line is the wrong shape — whatever R^2 says.

WHAT THE PATTERN MEANT

An R² of 0.75, and the line is still wrong



Studying helps a lot at first and then levels off. A straight line cannot level off, so it under-predicts at both ends and over-predicts in the middle — exactly the pattern in the errors.

SECTION 5 OF 5

Doing it for real

Sixty students, three lines of code, and one trap.

- scikit-learn
- Split first
- Reading the output
- Never extrapolate

SCIKIT-LEARN

The entire thing, in three lines

```
from sklearn.linear_model import LinearRegression

model = LinearRegression().fit(hours, marks)
print(model.coef_[0], model.intercept_)      #  8.0   26.0
```

.fit(X, y)

does all of it — the formula from slide 14, in one call

.coef_

the slope. One per input column, so it is always a list

.intercept_

the intercept. A single number

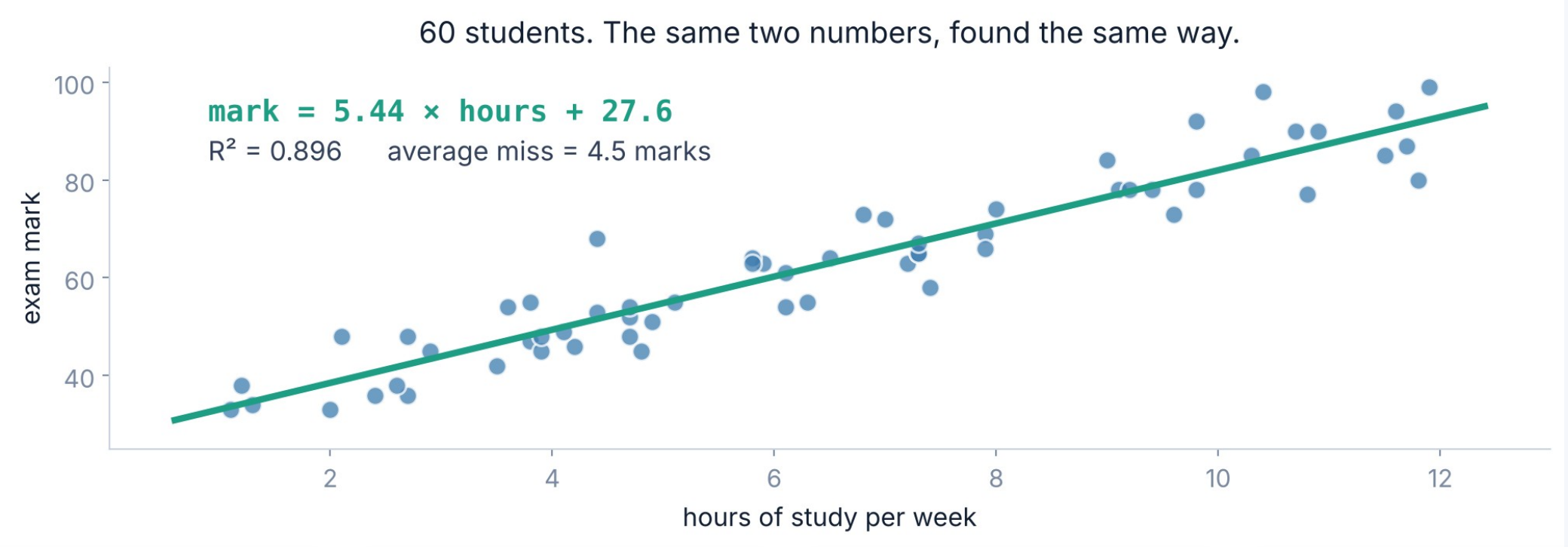
.predict(X)

run the rule on new students

It returns 8.0 and 26.0 — the same two numbers you worked out by hand. The computer is not doing anything you have not already done on paper.

A REAL-SIZED DATASET

Now sixty students instead of five



5.44
marks per extra hour

0.896
R^2 on the rows it studied

4.5
average miss, in marks

60
students

THE RULE FROM DAY 31, AGAIN

Hide some students before you fit

EXAMPLE

Fit on 70% of the students. Score on the 30% the model has never seen.

```
a, b, c, d = train_test_split(X, y, test_size=0.3, random_state=0)
model = LinearRegression().fit(a, c)
print(model.score(a, c), model.score(b, d))       #  0.900   0.883
```

0.900

on the rows it studied

0.883

on rows it had never seen

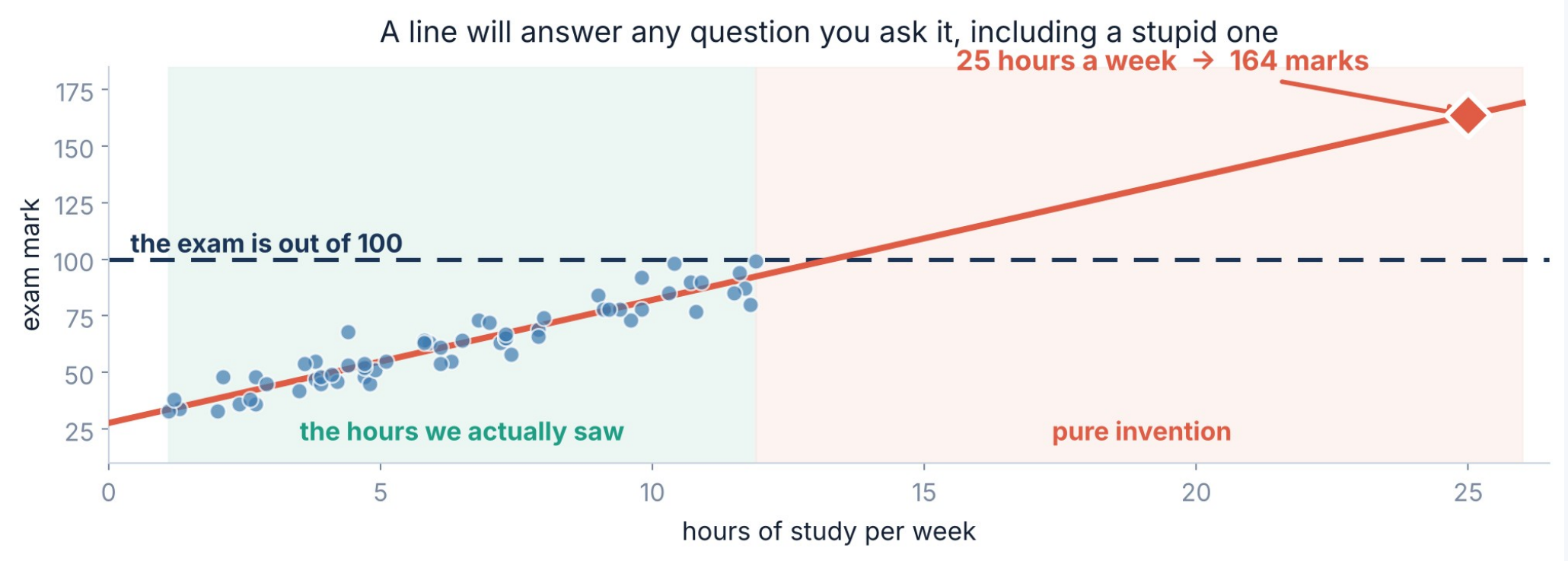
0.017

the gap between them

A gap of 0.017 is nothing. A straight line has only two numbers to adjust, so it is almost impossible for it to memorise — the opposite of the tree on Day 31 that scored a perfect 1.000 and then collapsed.

THE TRAP

A line will answer a stupid question without blinking



Nobody in the data studied more than 11.9 hours. Ask about 25 and the line says 164 marks on a paper out of 100. The line has no idea the exam has a ceiling — or that a week has 168 hours.

JUDGEMENT

When is a straight line the right choice?

USE A LINE WHEN

- the scatter plot genuinely looks like a band going up or down
- you need to EXPLAIN the result to somebody
- you have few rows
- you want a baseline to beat

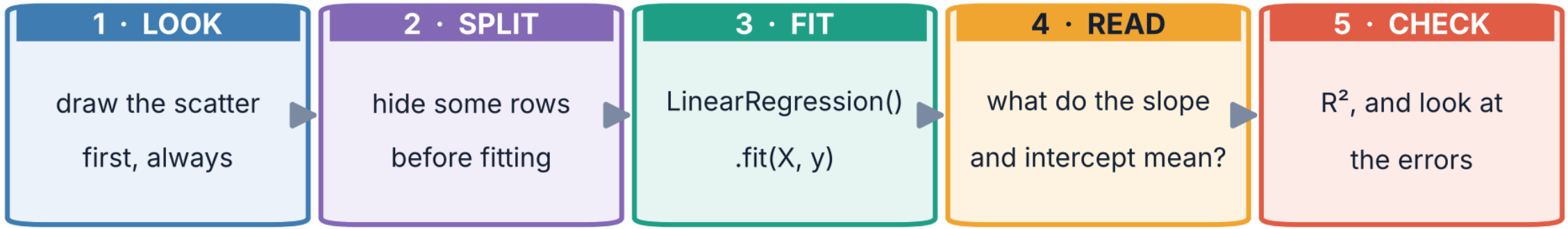
DO NOT USE A LINE WHEN

- the scatter plot curves or levels off
- the spread grows as x grows
- there are a few extreme outliers dragging it
- the real pattern is a step, not a slope

Always fit the line first anyway. It takes ten seconds and it gives you a number every fancier model has to beat. A gradient-boosted forest that scores 0.62 against a straight line's 0.60 was not worth the afternoon.

PUT IT TOGETHER

The five steps, every time



Step 1 is the one people skip. Draw the scatter plot before you fit anything. Ten seconds of looking will tell you whether a straight line is even the right shape — and will save you from reporting an R^2 of 0.75 on data that curves.

BEFORE YOU GO

Five mistakes everybody makes once

#	the mistake	what you will see	the fix
1	Fitting without plotting first	A respectable R^2 on data that curves	Draw the scatter. Always
2	Quoting the intercept	“Zero hours gives 26 marks”	Check 0 is inside your data
3	Predicting outside the data	164 marks out of 100	Note your min and max, and stay inside
4	Trusting R^2 on its own	A pattern hiding in the errors	Plot the errors against the input
5	Saying hours CAUSE marks	A confident, unprovable claim	A line measures movement, not cause

Number 5 is the one that gets people into print. Students who study more may also sleep better, attend more, and start earlier — the line cannot separate any of that.

SUMMARY

What to remember from today

A line is two numbers	slope and intercept. That is the entire model — nothing else is stored.
“Best” had to be defined	Adding errors gives zero for the worst line imaginable. Squaring them first is what makes the score mean something.
Least squares, literally	Every error becomes a square on the chart. The best line is the one with the smallest total area: 14, not 104.
Two routes, one answer	A formula gives slope 8 in one calculation. Gradient descent walks there in 800 steps. Only the second survives into big models.
R^2 compares with the average	0.979 means the line removed 97.9% of the error that guessing 50 every time would have left. It does not mean the line is the right shape.
Look at the errors	A pattern in the errors beats any score. And never ask the line about hours it has never seen.

19 QUESTIONS

Tonight's practice

A **Work it out** — 8 questions

Find the best line by hand, twice, with no computer

B **Find the mistake** — 4 questions

Four programs: wrong shape, bad extrapolation, leaked split

C **Write the code** — 5 tasks

Fit, read, score and diagnose a line on real data

D **Challenge** — 2 tasks

Build least squares from scratch, then catch a line that lies

Part D hands you a model with an R^2 of 0.93 and asks you to prove that its headline claim is worthless.

$\text{mark} = 8 \times \text{hours} + 26$

Two numbers, found by hand, confirmed by the computer. Every model you meet for the rest of this course is a bigger version of exactly this.

NEXT · DAY 34

Multiple linear regression — what happens when you have more than one input column.

Slides & notebooks by **Srinivasa Sai Chava** · Boston University